Capstone controlled experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Capstone controlled experiments: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Trace matched two-update learning-rate experiments

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Capstone hypothesis and baseline

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Look for: Write a testable capstone hypothesis and a credible baseline

Reference example, not the only acceptable wording: Teacher explanation: Predict max(0, a*x) without bias; baseline a = 1; candidate starts at 1, learning rate 0.1, two full-batch updates on T1/T2.

Hypothesis: candidate validation MSE on V1/V2 is strictly below the baseline MSE.

Efficiency and resource measurement

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate throughput and amortised time per example from repeated timings

Reference example, not the only acceptable wording: Teacher explanation: A: 2.0 seconds, 20 ms per example, 50 examples per second. B: 1.2 seconds, 12 ms per example, about 83.33 examples per second.

The boundary covers 100 completed sequential **prediction**s on pre**process**ed **input**s, excluding loading and pre**process**ing. It does not establish worst-case or percentile request latency.

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a reused feature extractor and a newly fitted **output** head

Reference example, not the only acceptable wording: Teacher explanation: The extractor gives h = 6 and the head predicts 1.

Weights [1, 1] and bias 0 stay frozen; only the threshold is fitted. h is a nonnegative sum of counts, with no probability mapping or calibration.

For the compact alternative schedule, use two 45-minute sessions. This capstone uses constructed scalar ReLU regression so every calculation is inspectable. It is an experiment in **training** and **evidence**, not a real-world deployment.

Preserve written plans, data roles and raw **prediction**s. Optional local Python requires no external packages or AI account. Do not expose independent-task solutions before collecting work.

Outcomes

Trace matched two-update learning-rate experiments

Select a candidate using a prespecified validation rule

Explain the limits of a one-factor comparison

Preparation

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Fictional data only; reserve the independent task and fresh retry until assessment

Prerequisite answer

Data and splits, initial parameters, architecture, loss, update count and evaluation procedure must stay fixed; change only the intended learning rate.

End goal for the pictorial case

Explain capstone controlled experiments with a traceable toy-network case and a bounded conclusion.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain and model the mechanism

> Model Capstone controlled experiments using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Capstone controlled experiments · 1

Continue with **training** (1,2) and (2,4), validation (3,6) and (4,8), and **prediction** max(0, a*x). Compare learning rates 0.1 and 0.2. Each run starts at a = 1, has no bias and takes two full-batch updates.

Keep the loss, data order and arithmetic fixed. This toy model has no randomness; repeating the exact run checks replay, not variation across seeds.

Record the comparison as an explicit revision to the one-candidate Week 33 plan before viewing new results. Validation may guide selection; sealed final data must not.

Select lower validation MSE, with a tie going to learning rate 0.1.

Explain Capstone controlled experiments · 2

For positive weights and these positive **input**s, the ReLU is active. **Training** MSE is ((a-2)^2 + (2*a-4)^2)/2 = 2.5*(a-2)^2. Its gradient is 5*(a-2). Update a to a - rate*5*(a-2). At rate 0.1, weights are 1.5 then 1.75.

At rate 0.2, weights are 2 then 2. The formula assumes the active region; do not extend it blindly to nonpositive weights.

Term	Meaning
Controlled comparison	A comparison holding specified conditions constant while changing an intended factor
Full batch	An update whose gradient uses all training rows
Selection rule	A rule for choosing a candidate using development evidence
Confounder	A differing condition that prevents isolating the intended explanation

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Same initial a = 1	Rate 0.1, two updates	Validation MSE 0.78125
Same initial a = 1	Rate 0.2, two updates	Validation MSE 0
Same validation rows	Choose lower MSE	Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Explain Capstone controlled experiments · 3

Guided: 2.5 - 0.3*2.5 = 1.75; validation MSE 0.78125. Equal final weights here give equal final **prediction**s, but the intermediate weights differ.

Exit: the score compares complete procedures with different budgets, not an isolated learning-rate effect; match update counts and all other settings.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 4

At a = 1.75, validation **prediction**s are 5.25 and 7. Errors -0.75 and -1 square to 0.5625 and 1, giving MSE 0.78125. At a = 2 the **prediction**s equal 6 and 8, so MSE is zero. Rate 0.2 wins the written validation rule.

This selection uses two validation cases, not four independent cases because two runs used them.

Model the reasoning aloud · 5

If one run used extra epochs or different **training** rows, the comparison would mix factors. Either rerun a matched experiment or report the differences and avoid claiming an isolated learning-rate effect.

A zero validation score is also compatible with a very simple task and says nothing about a new distribution.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

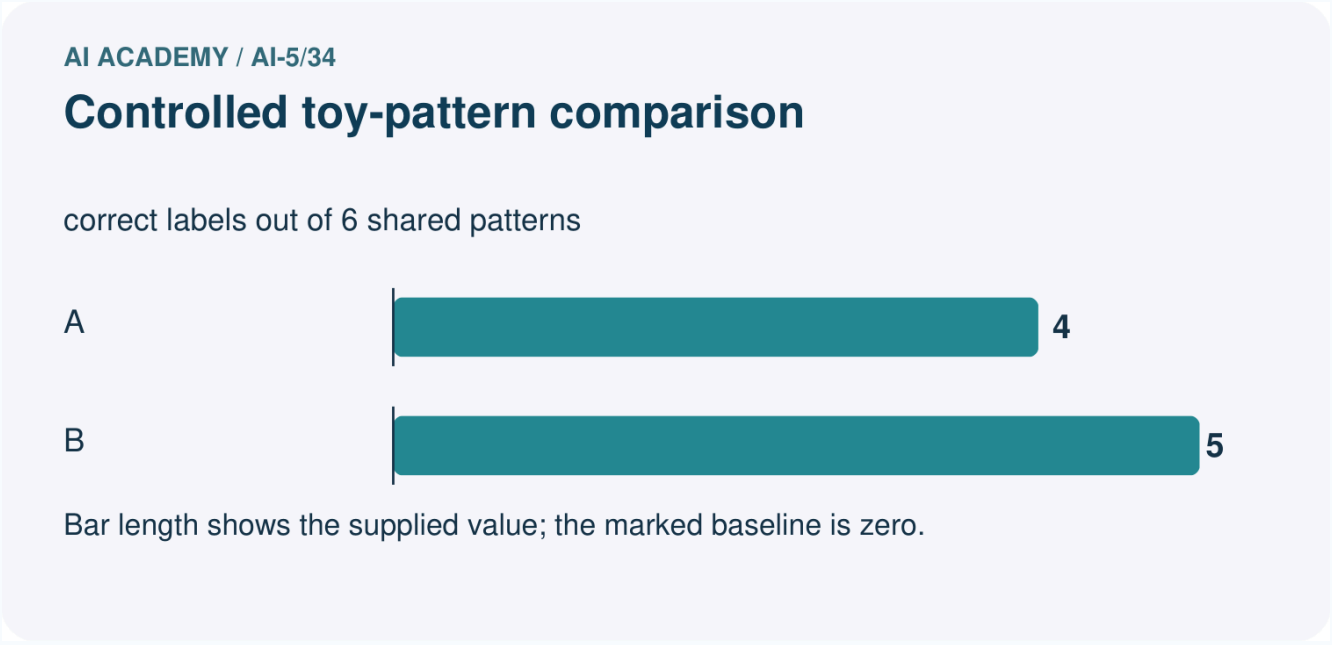
A pictorial worked case: Capstone controlled experiments

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

The facts we will use

Supplied facts

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/34

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



2. Count outcomes

A is 4/6 correct; B is 5/6.



3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the intervention	B adds the centre-feature connection; all other declared settings stay fixed.
2. Count outcomes	A is 4/6 correct; B is 5/6.
3. Inspect paired cases	Only case three changes correctness; case five fails for both.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

What this example does not establish

Keep the stated limit

These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

Change one thing in the pictured case

Change one thing

Change: Change B's **training** set as well as the feature connection.

Predicted result

Prediction: The experiment no longer isolates the connection change.

Why the result follows

Reason: A **training**-data difference supplies an alternative explanation for the observed result.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Capstone controlled experiments, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and ask students to explain the data roles.
7 to 18 minutes	Read section 1 and complete W1 with a recorded reason.
18 to 32 minutes	Trace section 2 and complete W2; compare raw values before conclusions.
32 to 45 minutes	Explain the diagram and complete W3 in pairs; record any support.
Session 2 start to 8 minutes	Complete the guided response and discuss why the evidence has limits.
8 to 20 minutes	Read the remaining explanation and prepare a concise experiment record.
20 to 35 minutes	Collect W4 independently. Require calculations and a bounded claim.
35 to 45 minutes	Use W5 and the exit response for feedback. Administer the fresh retry after feedback without the earlier solution.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Reduce help deliberately

Completion practice

Completion task: Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

Reasoning to listen for

Teacher check: Teacher explanation: Rate 0.1: 1 -> 1.5 -> 1.75. Rate 0.2: 1 -> 2 -> 2.

Gradients are -5 then -2.5 for the first run and -5 then 0 for the second.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Reusing two rows across runs leaves two distinct validation cases.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Compute the two correct counts and identify the changed case. Why would changing B’s **training** data weaken the conclusion about its added connection?

Reasoned teacher answer

A has four correct, B five; case three explains the count gain and case five fails for both.

Changing **training** data alongside the connection entangles two interventions, so the gain cannot be attributed to the connection alone.

Changed-case reasoning: The experiment no longer isolates the connection change. A **training**-data difference supplies an alternative explanation for the observed result.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Capstone controlled experiments investigation and use the first unsupported step to choose a teaching response.

More runs mean more validation examples: Reusing two rows across runs leaves two distinct validation cases.

The largest rate must be best: The active-region update may overshoot; results depend on task, rate and budget.

Zero validation error proves the final result: Final rows remain unopened and the selected candidate has not yet been evaluated on them.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Capstone controlled experiments questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: 2.5 - 0.3*2.5 = 1.75; validation MSE 0.78125. Equal final weights here give equal final **prediction**s, but the intermediate weights differ.

Exit: the score compares complete procedures with different budgets, not an isolated learning-rate effect; match update counts and all other settings.

W1 Name the controls

> **Workbook question · W1**

List six settings held fixed and the single intended changed factor.

Teacher reasoning · W1

Teacher explanation: Hold **training** and validation rows, initial a, architecture/bias policy, loss, update count and evaluation metric fixed. Change learning rate.

W2 Trace both runs

Workbook question · W2

Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

Teacher reasoning · W2

Teacher explanation: Rate 0.1: 1 -> 1.5 -> 1.75. Rate 0.2: 1 -> 2 -> 2. Gradients are -5 then -2.5 for the first run and -5 then 0 for the second.

W3 Select without extra evidence

Workbook question · W3

Recompute validation MSE for the final weights and apply the written **selection rule**. How many distinct validation rows were used?

Teacher reasoning · W3

Teacher explanation: MSE 0.78125 for a = 1.75 and 0 for a = 2, so select rate 0.2. There are two distinct validation rows; re-evaluation does not add cases.

W4 Run a new comparison independently

Workbook question · W4

For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Teacher reasoning · W4

Teacher explanation: Rate 0.1: gradients -10,-5; weights 2,2.5. Validation errors -1.5,-2 give MSE 3.125. Rate 0.2: weights 3,3, errors 0,0 and MSE 0. Select 0.2, with only validation **evidence** on two constructed rows.

W5 Audit a mixed comparison

Workbook question · W5

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different **training** rows. Can the difference be attributed to rate alone? Propose a repair.

Teacher reasoning · W5

Teacher explanation: No: update count and **training** data also differ.

Rerun both rates with the same initial parameters, architecture, rows, loss, two-update budget and evaluation procedure; preserve and label the earlier mixed comparison.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires weights 2.5 and 3, MSE 3.125 and 0, selection 0.2 and a validation-only conclusion. Fresh retry: targets are 4*x on the same **input**s, gradient 5*(a-4). Final weights are 3.25 and 4; validation MSE 7.03125 and 0.

Require the same data-role limit.

Assess independent transfer

Independent question · withhold the key

Selected case: For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Rate 0.1: gradients -10,-5; weights 2,2.5. Validation errors -1.5,-2 give MSE 3.125. Rate 0.2: weights 3,3, errors 0,0 and MSE 0. Select 0.2, with only validation **evidence** on two constructed rows.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Capstone controlled experiments to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1.

Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Week 35 evaluates a frozen candidate and records limitations before considering revision.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Capstone controlled experiments lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Trace matched two-update learning-rate experiments

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

Supported practice answer

Reasoned practice answer: Teacher explanation: Rate 0.1: 1 -> 1.5 -> 1.75. Rate 0.2: 1 -> 2 -> 2. Gradients are -5 then -2.5 for the first run and -5 then 0 for the second.

Check every learner before discussion: Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Reusing two rows across runs leaves two distinct validation cases.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change B's **training** set as well as the feature connection. Require a **prediction** and a reason before comparing. Expected result: The experiment no longer isolates the connection change.

Reason: A **training**-data difference supplies an alternative explanation for the observed result.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

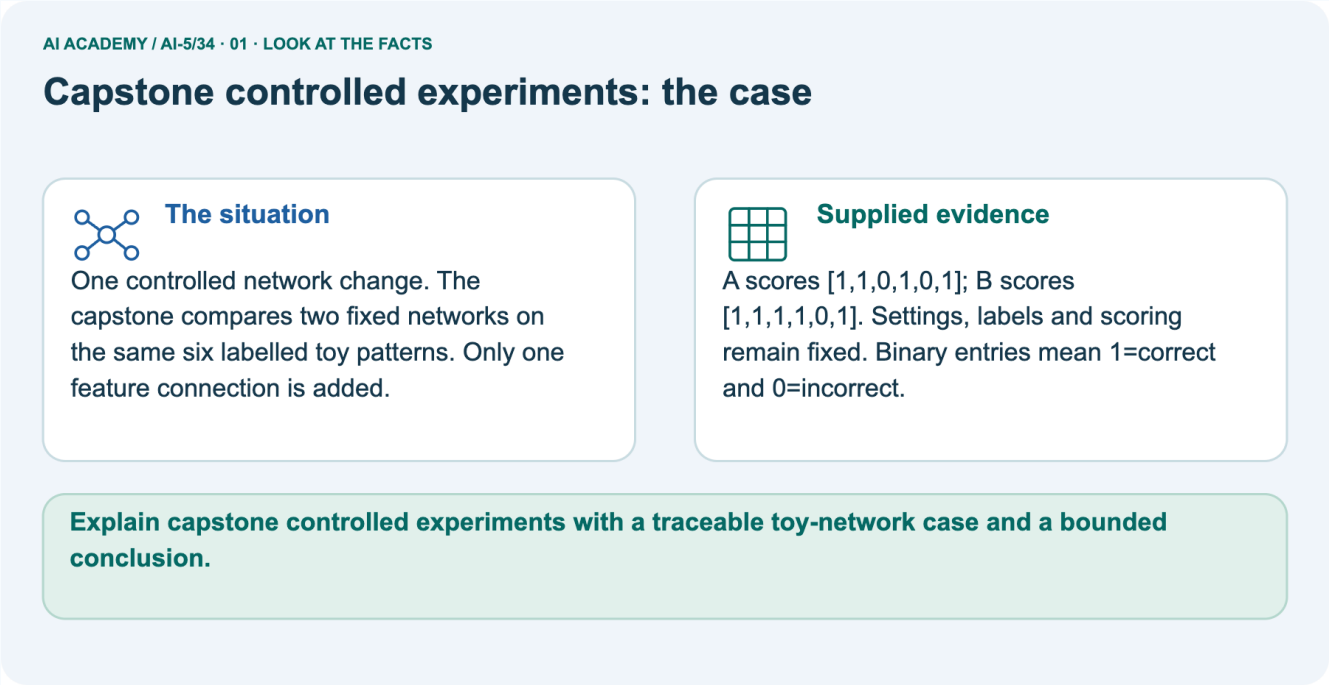
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Capstone controlled experiments graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone controlled experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added. Supplied facts: A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1].

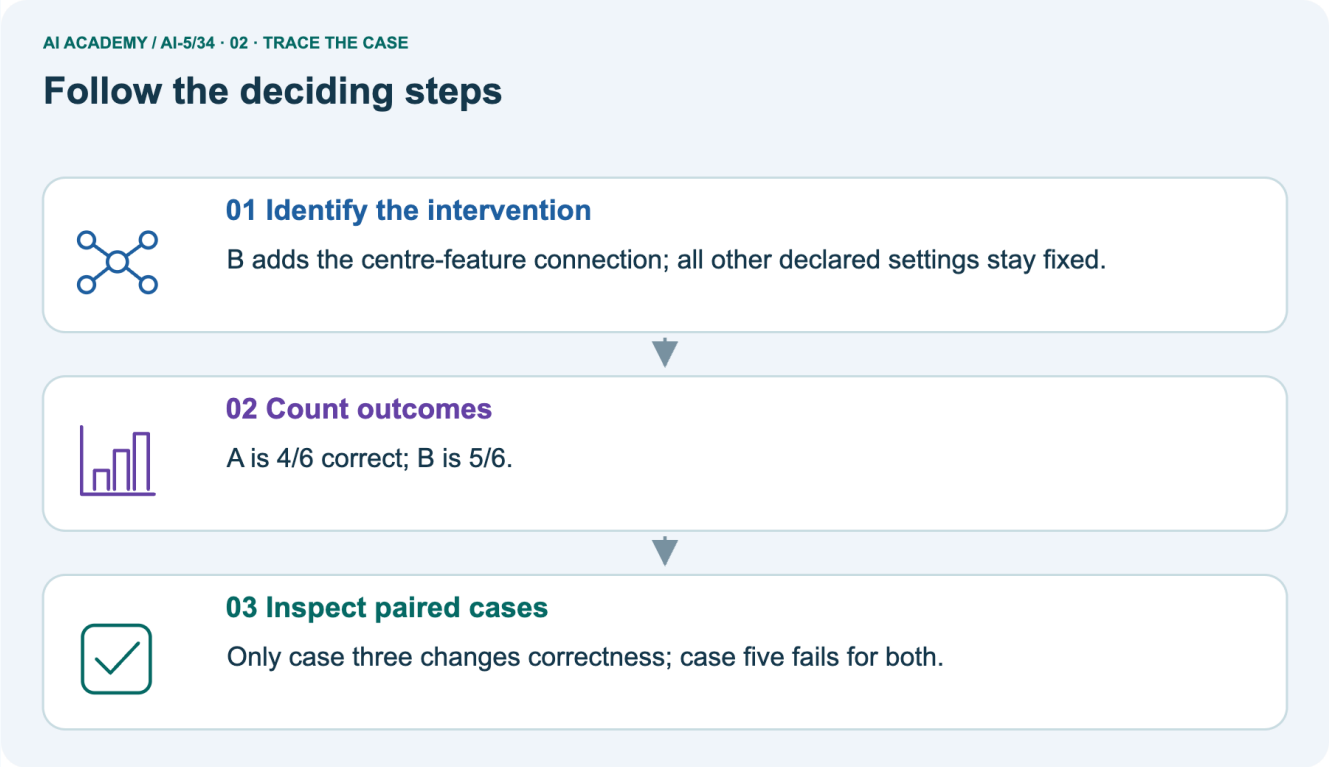
Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

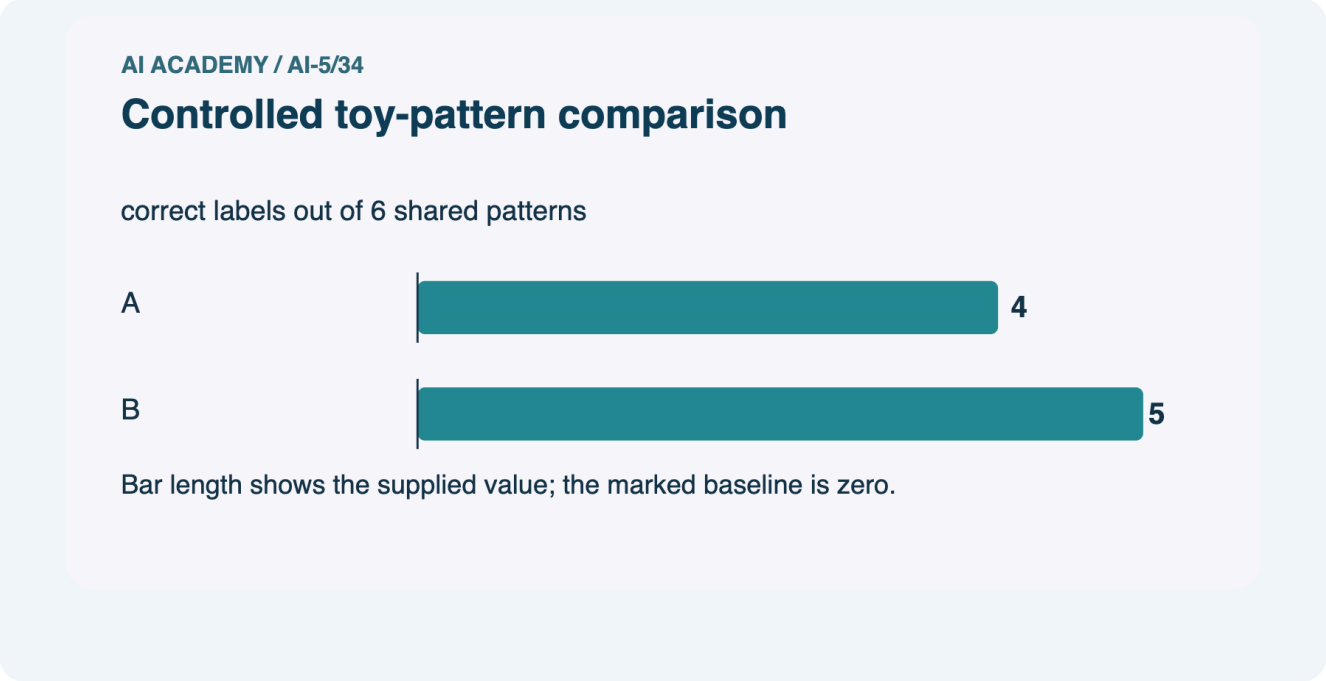
Identify the intervention: B adds the centre-feature connection; all other declared settings stay fixed. → Count outcomes: A is 4/6 correct; B is 5/6. → Inspect paired cases: Only case three changes correctness; case five fails for both.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Controlled toy-pattern comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled toy-pattern comparison. Units: correct labels out of 6 shared patterns. A: 4. B: 5.

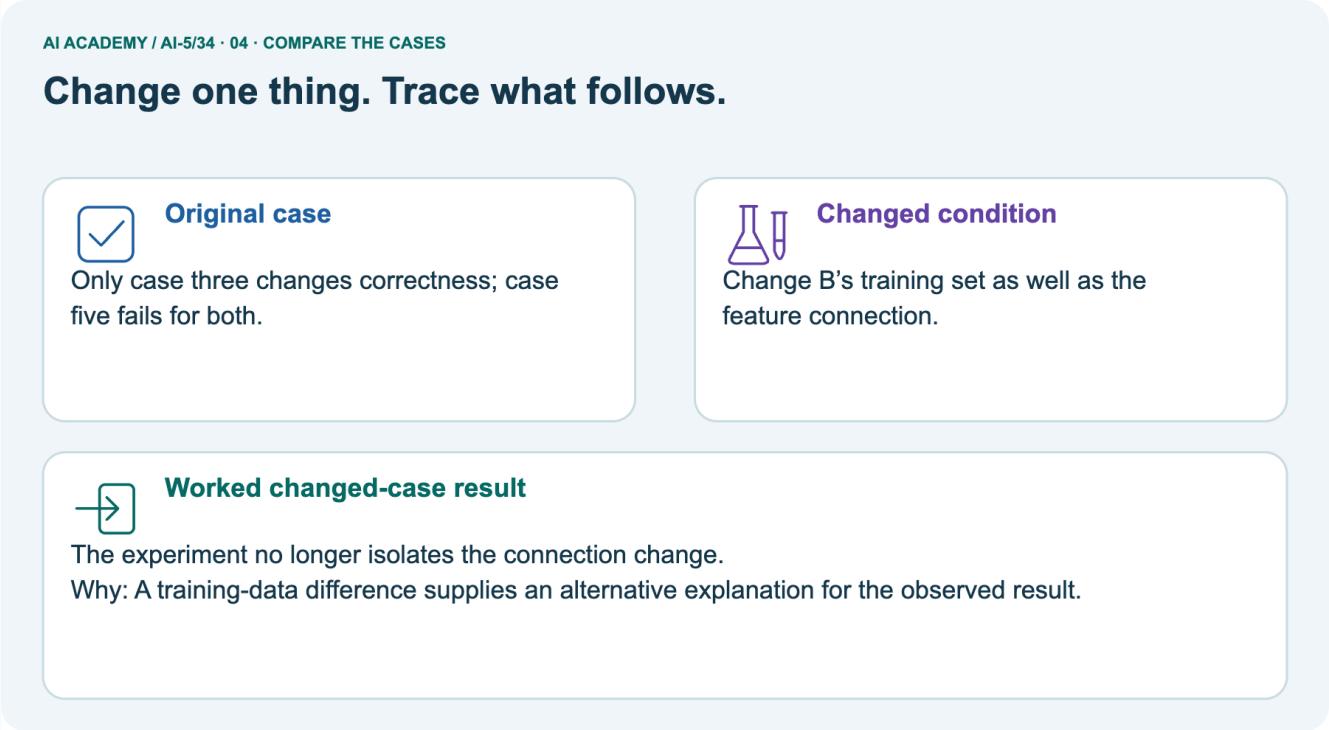
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only case three changes correctness; case five fails for both. Changed condition: Change B's **training** set as well as the feature connection.

Expected result: The experiment no longer isolates the connection change. Reason: A **training**-data difference supplies an alternative explanation for the observed result.

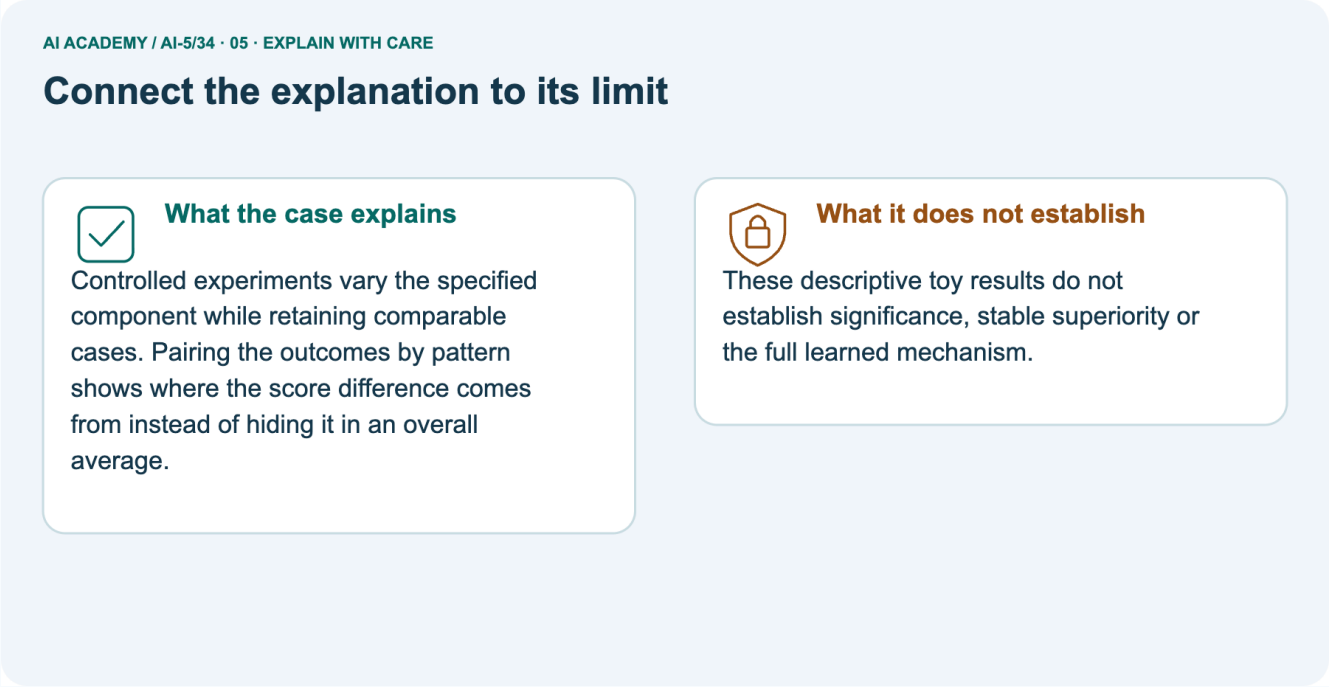
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A **training**-data difference supplies an alternative explanation for the observed result.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Limit: These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

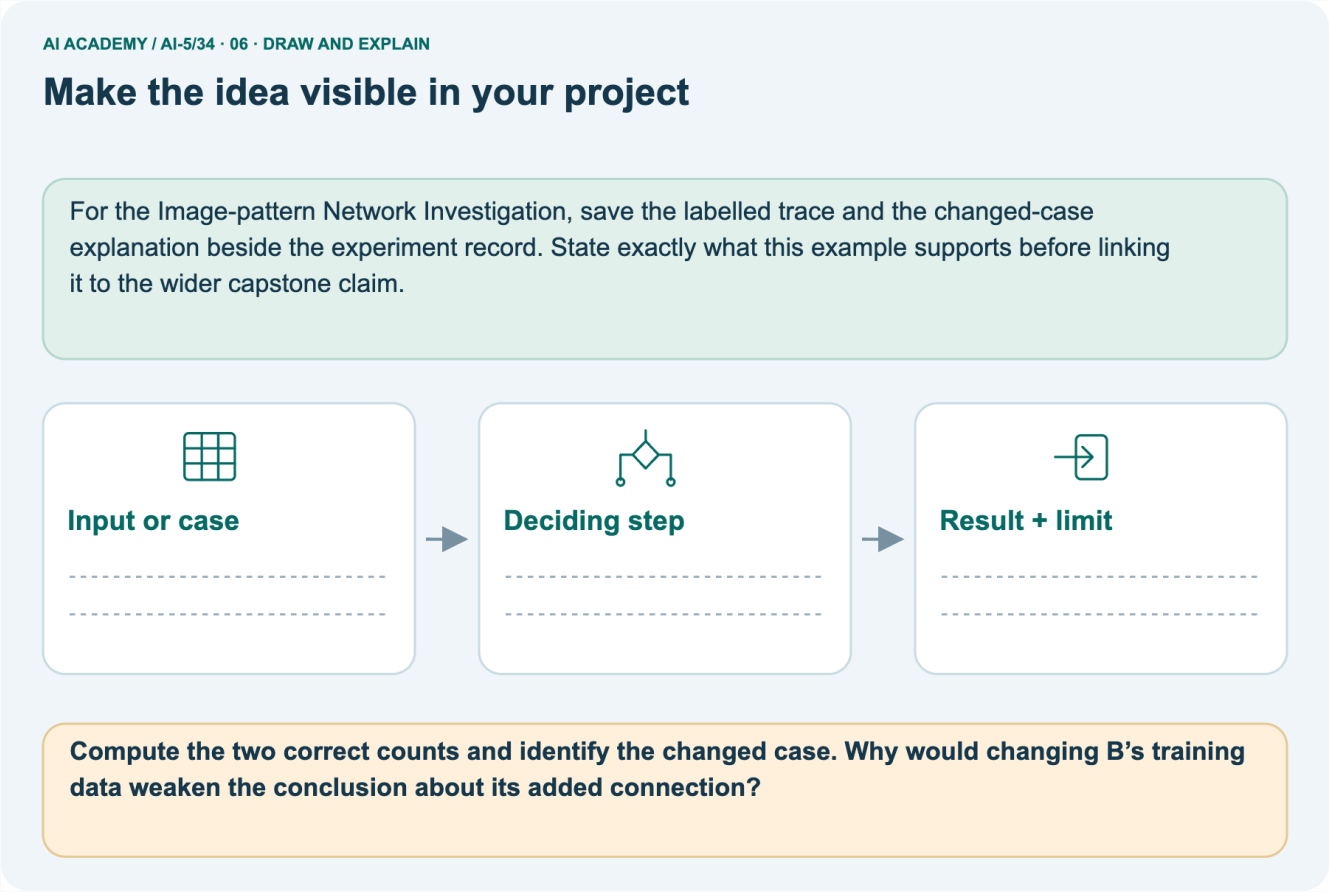
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A has four correct, B five; case three explains the count gain and case five fails for both.

Changing **training** data alongside the connection entangles two interventions, so the gain cannot be attributed to the connection alone.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.